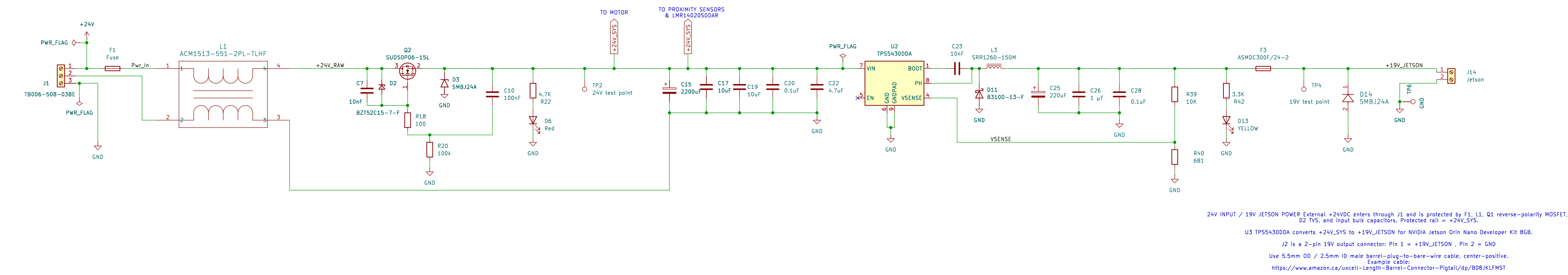
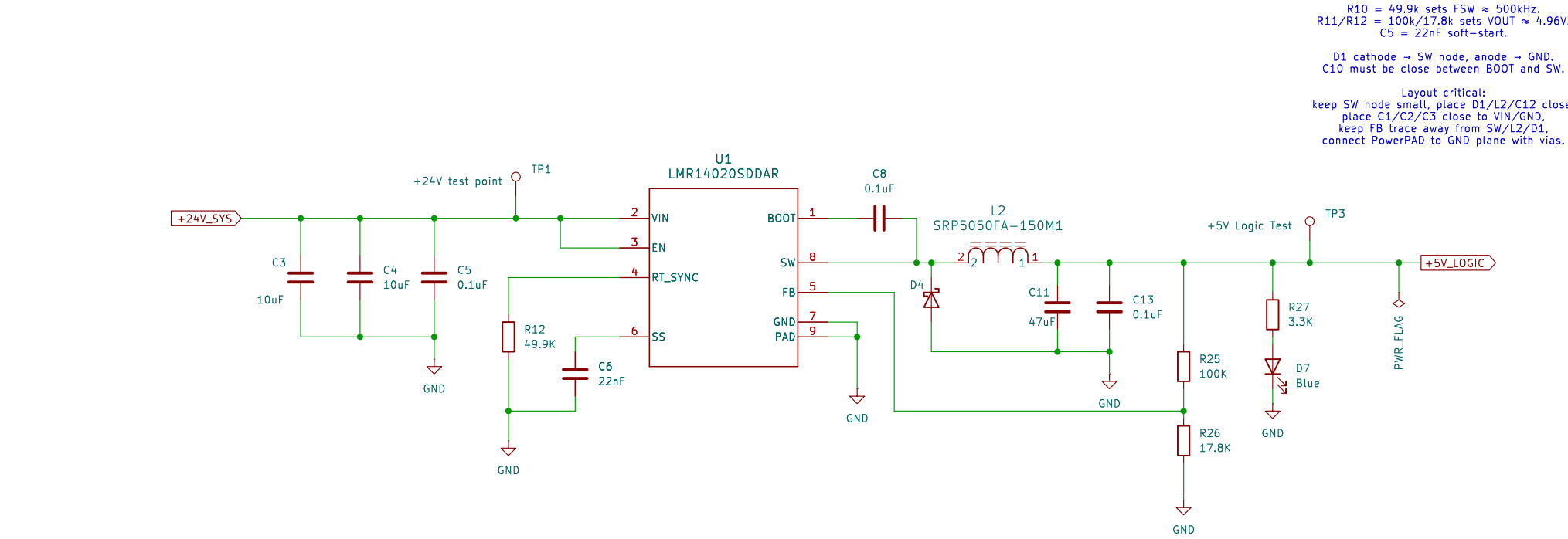


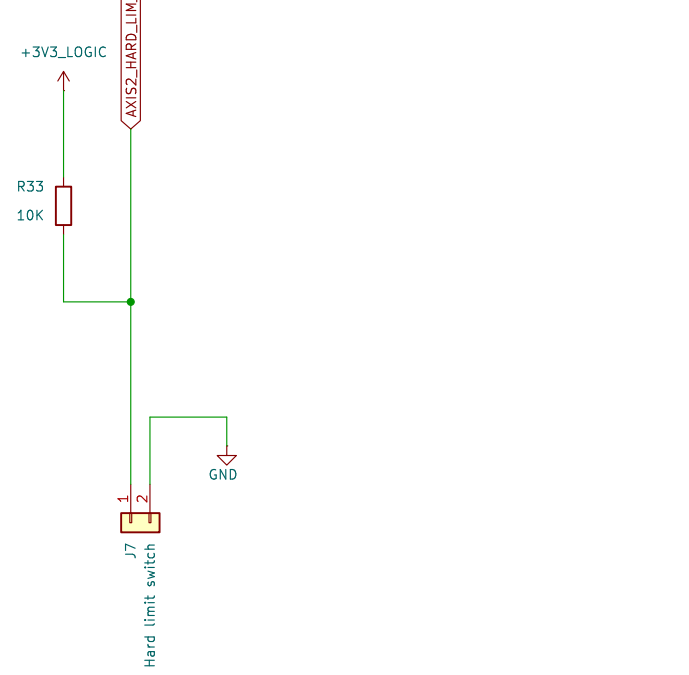
24VDC INPUT / 19VDC



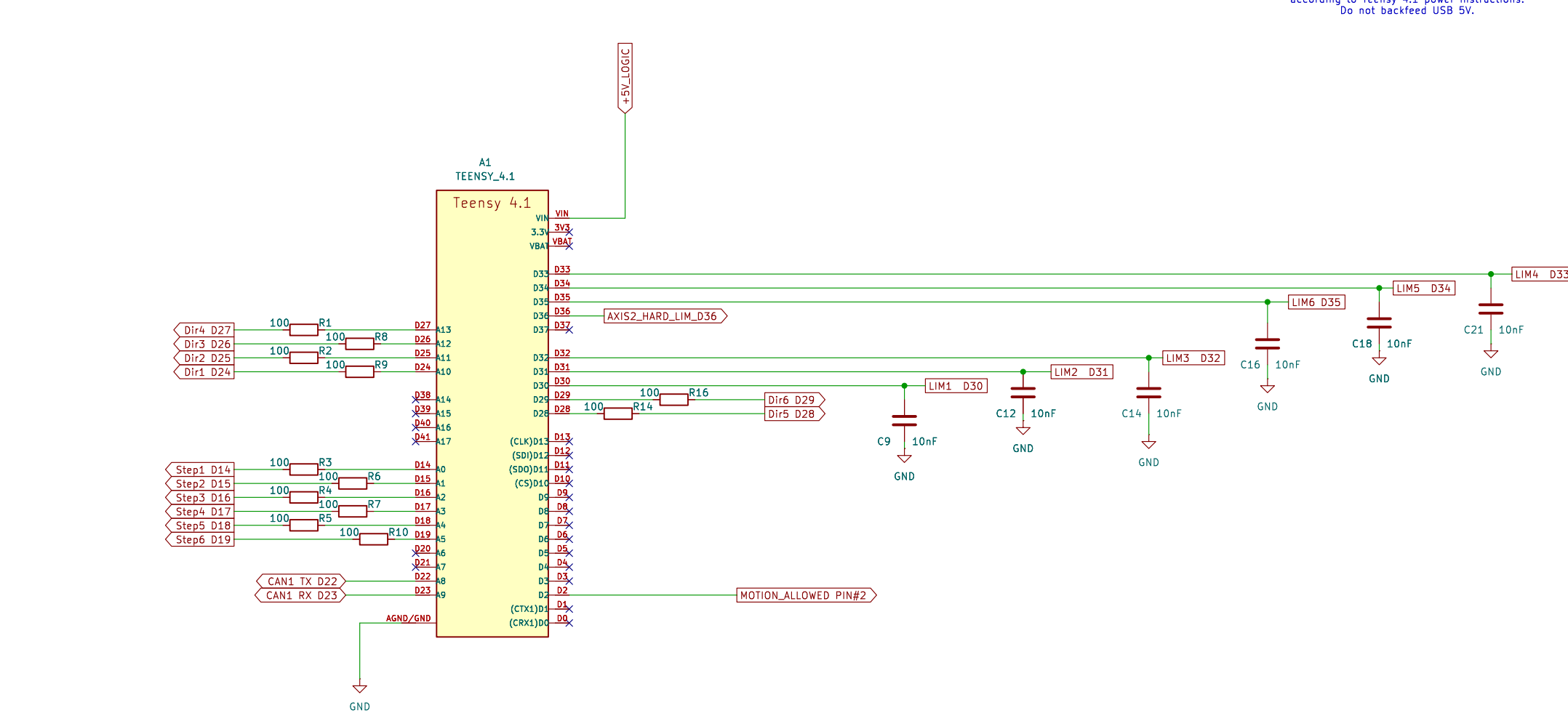
+24VDC 5VDC BUCK



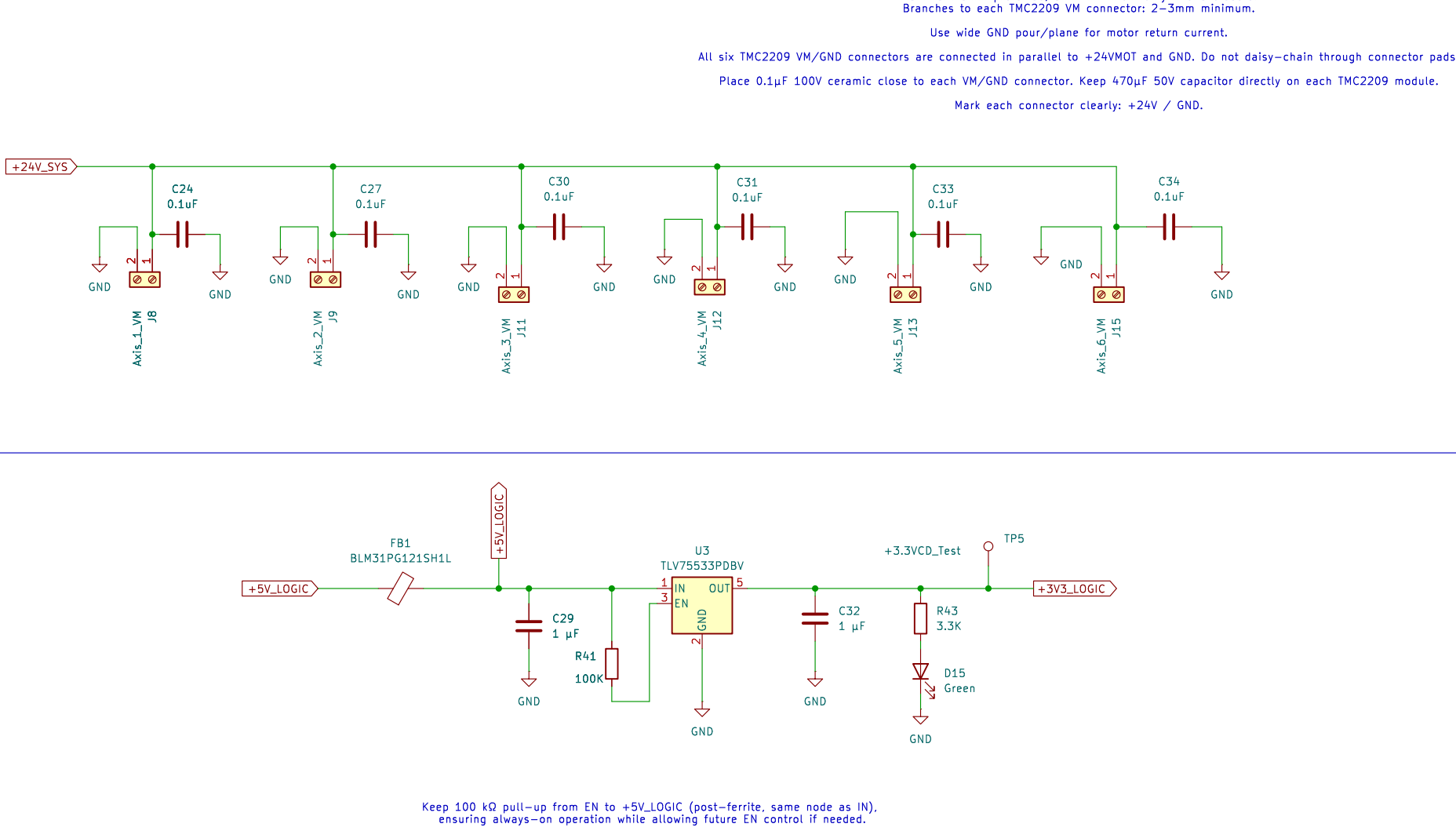
Hard limit switch



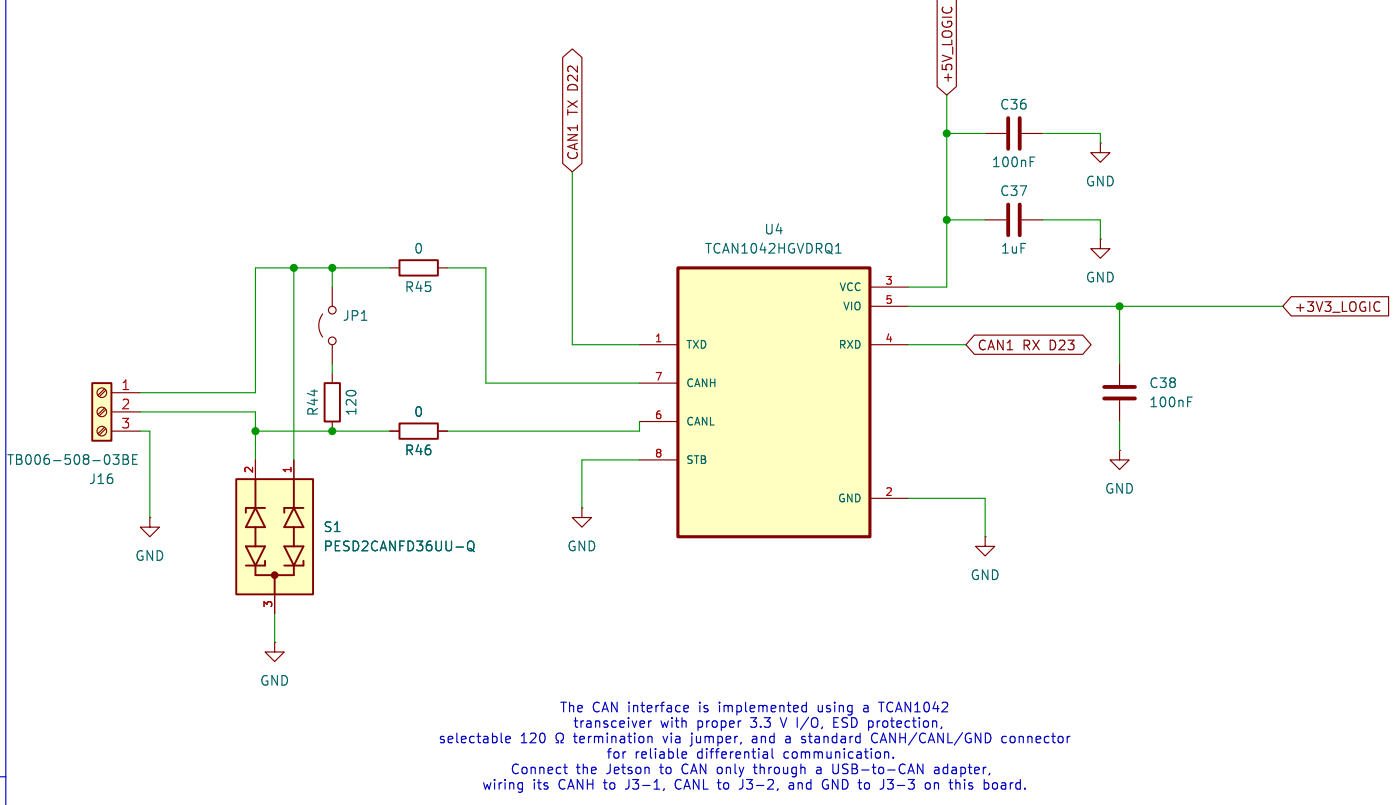
TEENSY_4.1



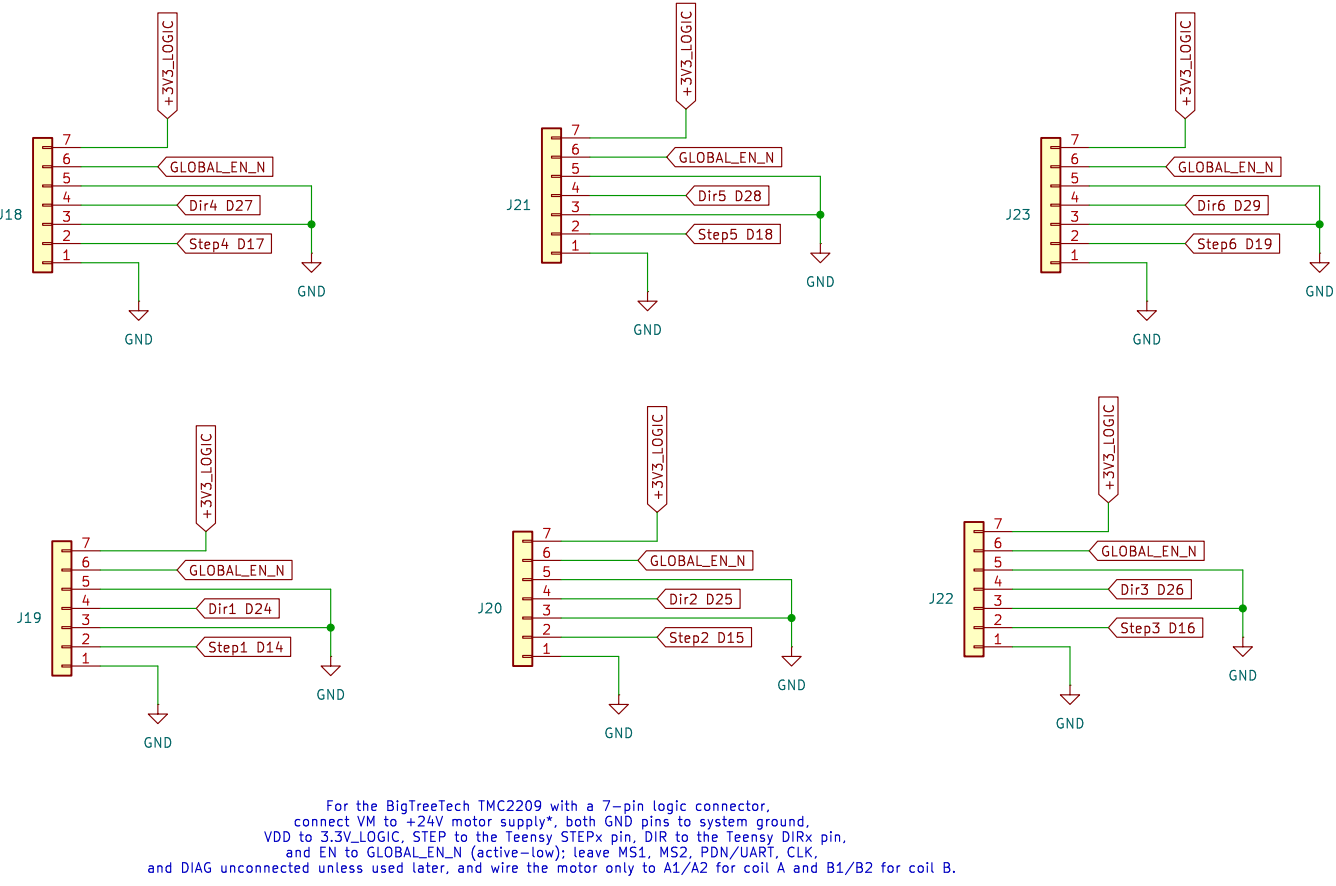
+24VMOT Motor Power Supply



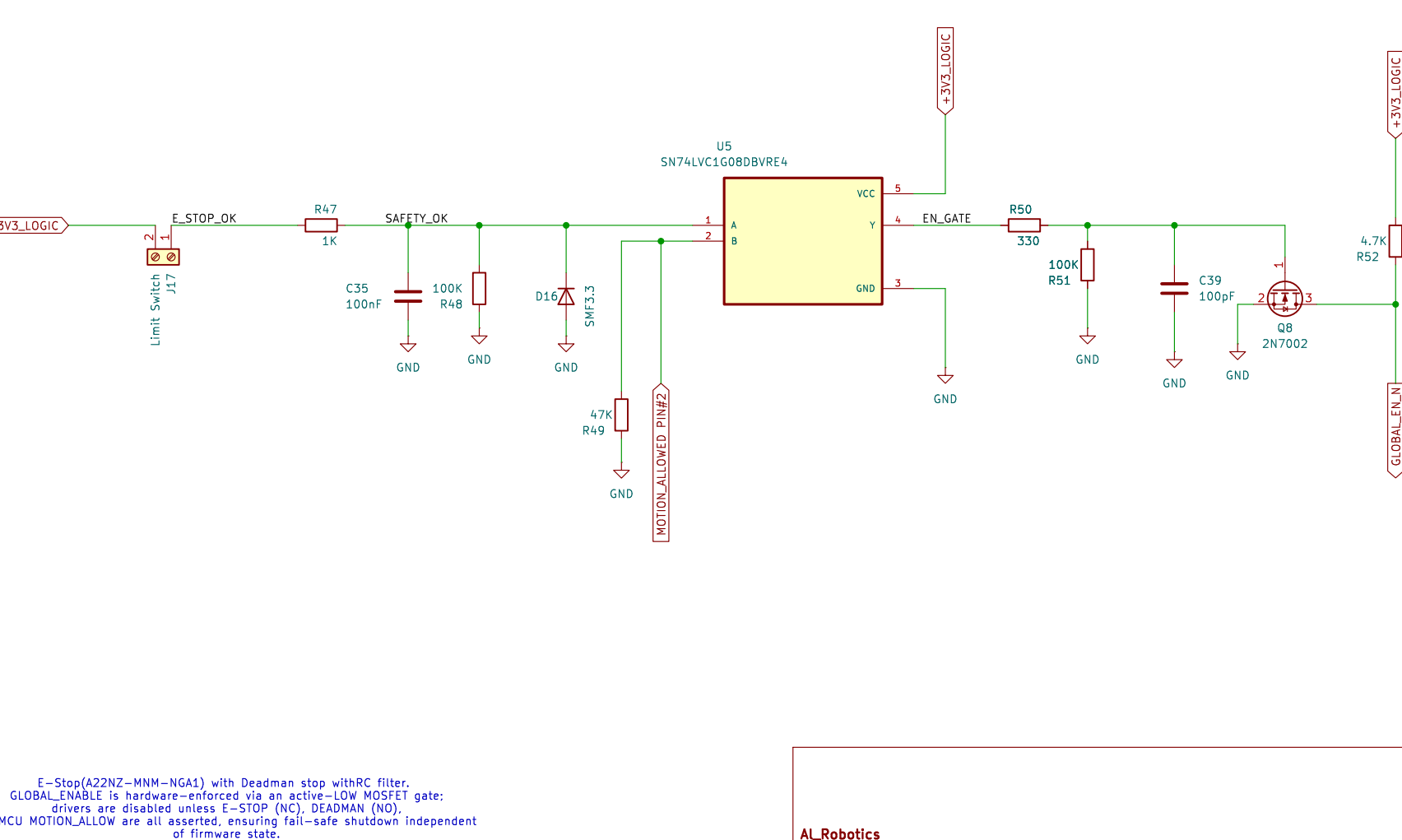
CAN Interface



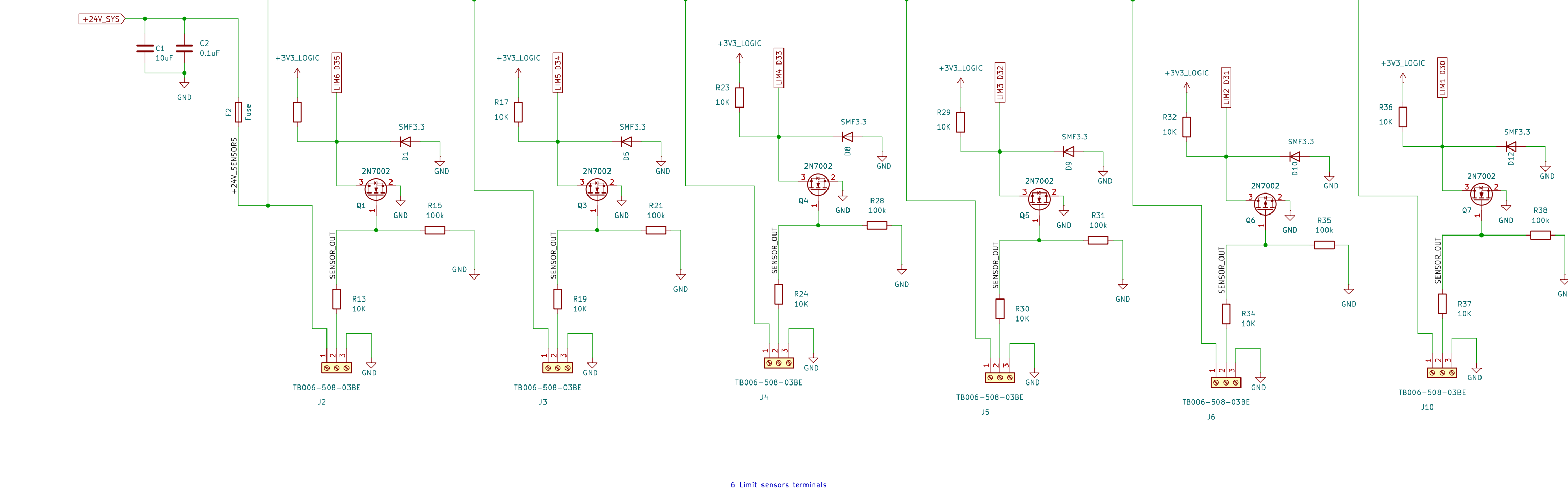
TMC2209 CONNECTORS



E-Stop



PROXIMITY SENSOR



SCHEMATIC NOTE - 24V PWM PROXIMITY SENSOR INPUTS
Sensor connector: TB006-50B-03BE
LIBA3-2-2-AY sensor wiring:
Brown wire - Pin 1 = +24V_SENSORS
Black wire - Pin 2 = SENSOR_OUT
Blue wire - Pin 3 = GND
+24V_SENSORS is derived from +24V_SYS through F5 retableable PTE fuse.
Each SENSOR_OUT signal is converted to Teensy 3.3V logic using a 2N7002 MOSFET interface:
SENSOR_OUT / black wire - 10K resistor - 2N7002 gate
2N7002 gate - 100K pull-down - GND
2N7002 source - Teensy LIM Input
Teensy LIM Input - 10K pull-up - +3.3V
2N7002 drain - 100K pull-down - GND
2N7002 protection diode is placed on the Teensy LIM input side only:
Cathode/stripes - LIM Input
Anode - GND
Expected logic:
No metal target = LIM LOW
Metal detected = LIM HIGH